

K-Field Three-Element Plate Thruster

Strict linear-inertia realization using three phase-balanced two-axis planar masses

Three matched 100 mm x 100 mm x 1 mm plate modules
Same circular chirality; carrier phases chosen to cancel rigid-body force and torque

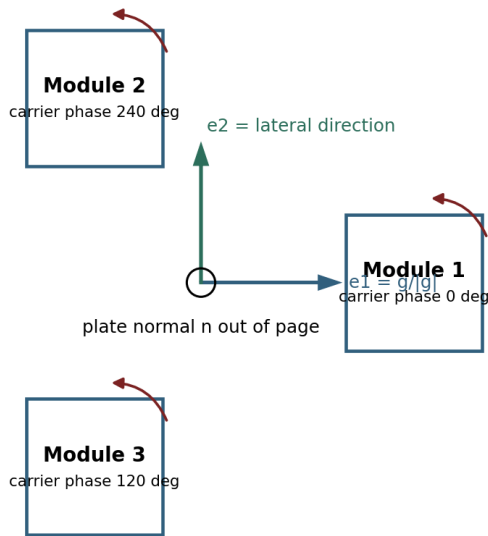


Figure 1. Three matched two-axis plate modules. Each plate executes the same in-plane circular translation. The global carrier phases are separated by 120 degrees; the spatially counter-wound assignment shown cancels ideal first-harmonic force and moment while preserving common chirality.

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Archival foundational education reference | Publication epoch 1785012327

Generated UTC 2026-07-25T20:45:27Z

Companion machine-readable file: K-Field_3_element_plate_thruster_1785012327.json

Document Contract and Abstract

Field	Value
Document class	Archival foundational education reference
Coordinate frame	ICRS Cartesian for field vectors; nanometres for Cell3 physical geometry; metres for the three-plate device
Principal investigator	P. Dwayne Esterline, Krytronx, LLC
Classification	Proprietary Confidential Information and Trade Secret of Krytronx, LLC. All Rights Reserved. Do Not Distribute.
Numerical source policy	All source constants, Cell3 nodes, matrices, and plane families are carried from the supplied strict linear-inertia machine record. Device figures and performance values are deterministic evaluations of those equations and the stated 100 mm x 100 mm x 1 mm geometry.
Authoritative source record	K-Field_linear_inertia_derivation_1784989727(1).json; SHA-256 4651a532fe43e8091748ef4f7e28bb899e1a2ea798d689ba1724e99530cd55ab
Force-status policy	Forces and accelerations are calculated model outputs. The report does not present a newly measured force record.
Field coupling	The strict law uses kappa_F directly. No unknown H or other field-transfer multiplier is introduced.
PDF generation	Python ReportLab; built-in core fonts only; dynamic table widths; Paragraph wrapping in every table cell; repeated table headers.
Companion artifact	K-Field_3_element_plate_thruster_1785012327.json

This paper develops a direct three-module realization of the strict linear-inertia phased-array result. Each active mass is a planar plate measuring 100 mm by 100 mm by 1 mm. Two orthogonal in-plane actuators drive the plate center of mass in quadrature, so the plate translates around a circle while its orientation remains fixed. This two-coordinate motion generates the same leading antisymmetric configuration-space current that the earlier thirteen-axis construction synthesized from many one-dimensional coordinates. The reduction is exact at the level of the strict local composition tensor; the response depends on the oriented current $\Omega = \langle \mathbf{r} \times \mathbf{r} \dot{} \rangle$, not on the number of actuator axes used to create it.

Three identical modules are operated with global carrier phases separated by 120 degrees while preserving the same internal 90-degree quadrature and the same chirality. The ordinary first-harmonic reaction forces cancel because the three cube roots of unity sum to zero. The second-harmonic terms cancel because their phases are doubled to another complete three-phase set. The phase-invariant mean lateral contributions remain equal and add. For an equilateral physical placement, counter-winding the carrier phase against the placement azimuth cancels the ideal first-harmonic moment as well as the net force.

Using the supplied Earth-CMB state, the response vector $\mathbf{g} = \mathbf{G}_{\text{phase}} \mathbf{V}_E$ has magnitude 930825.217270435 m/s. The optimum plate normal is $\mathbf{n}_{\text{opt}} = (-0.315865036407, 0.646162683024, -0.694768354089)$, and the positive lateral direction is $\mathbf{e}_{\text{ll}} = (0.077469585422, 0.747375863362, 0.659869519070)$. The strict maximum acceleration law is $|\mathbf{a}_K| = 8.989366088879\text{e-}5 f C_{\text{pol}} \text{ m/s}^2$, with frequency f in hertz and measured circularity C_{pol} . No unknown transfer coefficient appears.

Reading and Status Conventions

Source-fixed quantity. A value copied from the supplied Cell3 linear-inertia record, including B, W, $\mathbf{G}_{\text{phase}}$, $\mathbf{V}_E$, $\mathbf{q}_{\text{dot}_E}$, $\mathbf{g}$, $\mathbf{kappa}_F$, the 27 nodes, and the three plane families.

Derived identity. A deterministic algebraic or numerical consequence of the source-fixed quantities, such as the orientation triad, three-phase cancellation, or force-frequency coefficient.

Design input. A physical parameter selected for this device, such as the plate dimensions, density design value, translation amplitude, physical spacing, or operating frequency.

Measured implementation quantity. A parameter to be obtained from the built mechanism, including effective moving mass, modal amplitude, phase error, mode matching, stiffness, damping, and temperature response.

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1. Complete Result in Plain Language

One two-axis plate is sufficient to generate the derived mean lateral term. The plate does not rotate about its own normal. Its orientation remains fixed while its center of mass translates in a circle. Two orthogonal linear coordinates, driven at equal frequency and 90 degrees of internal phase separation, produce a nonzero antisymmetric current. That current is the strict linear-inertia source variable.

The central change in description is that the active state is not rigid-body spin. It is an oriented phase current formed by two linear coordinates. This converts a rotational-looking response into a stationary-geometry device problem and opens a direct pathway to piezoelectric, flexural, acoustic, electrostatic, electromagnetic, and other phase-controlled implementations.

Three plates are used for mechanical balance, not because the K-field term requires three sources. Their carrier phases are 0, 120, and 240 degrees. Each retains the same x-to-y quadrature sign, so all three circular translations have the same chirality. The ordinary carrier forces cancel; the calculated mean lateral forces add. The ideal second-harmonic ripple also cancels.

Single plate: $r_k(t) = A [\cos(\omega t + \psi_k) e_1 + \sin(\omega t + \psi_k) e_2]$
 Three phases: $\psi_k = 0, 2\pi/3, 4\pi/3$
 Common chirality: y channel remains 90 degrees behind x in every module
 Mean result: $F_{K,total} = 3 m_p \pi \kappa_F f C_{pol} (n \times g)$

For the 100 mm by 100 mm by 1 mm plate and the design density 1600 kg/m³, each plate has volume 1.0e-5 m³ and mass 0.016 kg. The three-module active mass is 0.048 kg. At full circularity and optimum orientation, the total mean-force coefficient is 4.314895722662e-6 N/Hz. The strict equation contains no finite intrinsic optimum frequency; the implementable frequency is the highest one at which the three modules preserve matched rigid translation, stable quadrature, structural integrity, and the intended effective moving mass.

Essential phase distinction. Changing every module by a common carrier phase only moves the plates to different positions on their circles; it does not change chirality or the mean result. Reversing the internal x-to-y phase order changes the sign of the antisymmetric current and reverses the lateral force.

2. Source Geometry and Coordinate Lock

2.1 Cell3 physical basis and address map

Cell3 uses an ordered three-channel address $q = (q_a, q_b, q_c)$. Physical position is obtained from the supplied basis B through the centered half-step rule $p_L = B L / 2$. The 27-node local registry is the complete set L in $\{-1, 0, 1\}^3$. This report preserves the source ICRS ordering and nanometre scale.

$p_L = (1/2) B L$
 $\dot{q} = 2 B^{(-1)} v$
 $G_{phase} = 4 B^{(-T)} W B^{(-1)}$

Matrix	Row 1	Row 2	Row 3
B (nm), row 1	0.32750124207	0.035330775844	-0.010129317627
B (nm), row 2	-0.054897007886	0.207319850441	-0.389787702364
B (nm), row 3	-0.037487502896	-0.547401085392	-0.689734408363
W (nm ²), row 1	0.083353302633	0.137500576604	0.102559211222
W (nm ²), row 2	0.137500576604	0.692400042849	0.550800904453
W (nm ²), row 3	0.102559211222	0.550800904453	0.587747652353
G_phase, row 1	2.9764106378	1.2431507795	-2.0803600354
G_phase, row 2	1.2431507795	2.5922286635	-2.3178839937
G_phase, row 3	-2.0803600354	-2.3178839937	6.7735709133

Cell3 centered half-step registry: all 27 nodes at true relative scale

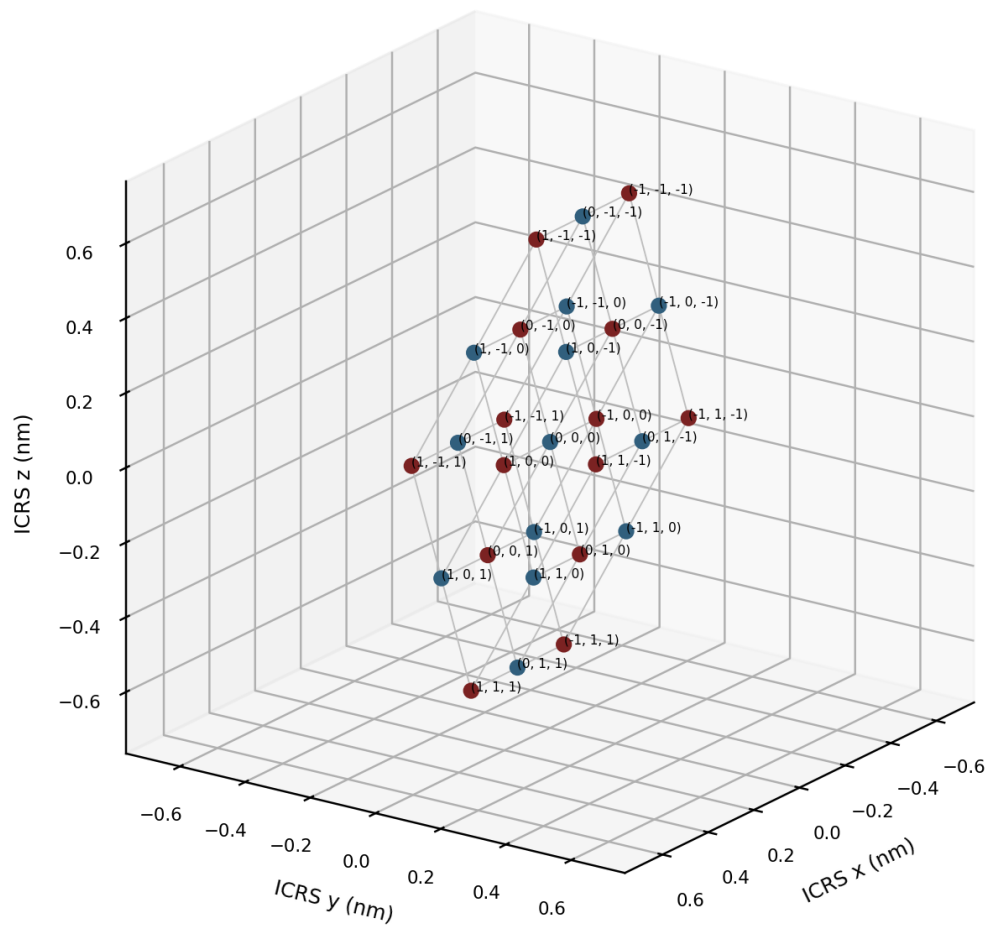


Figure 2. All 27 centered Cell3 nodes computed from $p_L = B L / 2$ for L in $\{-1,0,1\}^3$. Neighbor lines connect addresses differing by one unit in exactly one channel. Blue and maroon denote the supplied node polarities.

Cell3 orthographic projections with one physical scale

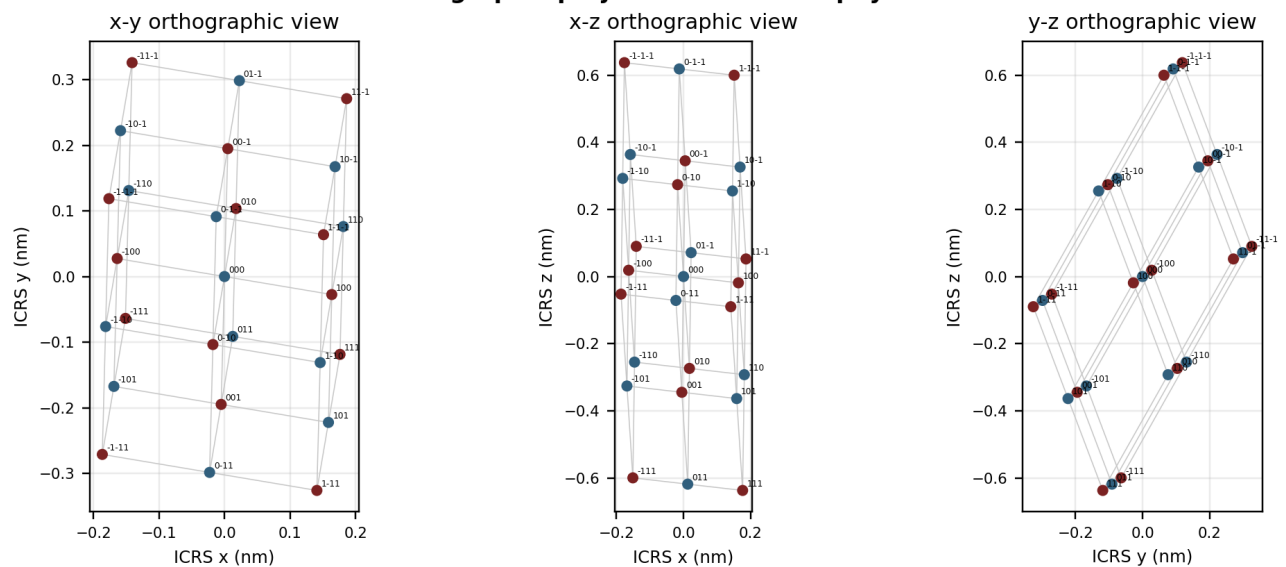


Figure 3. Orthographic x-y, x-z, and y-z projections of the same 27-node registry. Every panel uses equal physical scaling in nanometres, preventing perspective enlargement or axis distortion.

2.2 Base-cell dimensions

Quantity	Value	Unit
Cell volume	0.117914891916699	nm ³
Edge a	0.334179679073151	nm
Edge b	0.586410890414836	nm
Edge c	0.792319765037693	nm
Included angle a-b	83.933483012313	deg
Included angle a-c	80.448130018338	deg
Included angle b-c	50.363231813756	deg
Face area a-b	0.1948691654699	nm ²
Face area a-c	0.261106235799531	nm ²
Face area b-c	0.357809539128562	nm ²

The three edges and included angles define an oblique cell. Plane spacing must therefore be computed from the reciprocal geometry rather than inferred from a rectangular drawing. The plane-family normals and spacings below are source-fixed quantities.

2.3 Plane families and all centered planes

Channel	Parallel face	Unit normal in ICRS	Adjacent half-step spacing (nm)	Primitive boundary spacing (nm)	Face area (nm ²)
a	bc	(-0.99596519, 0.083602163, -0.032619291)	0.164773264855	0.329546529709	0.357809539129
b	ac	(0.089052594, 0.866576769, -0.491033948)	0.225798689862	0.451597379724	0.2611062358
c	ab	(0.194092201, 0.913177169, 0.358379235)	0.302548870757	0.605097741515	0.19486916547

Cell3 plane family a: fixed-a planes parallel to face bc
Adjacent centered planes = 0.164773264855 nm; primitive spacing = 0.329546529709 nm

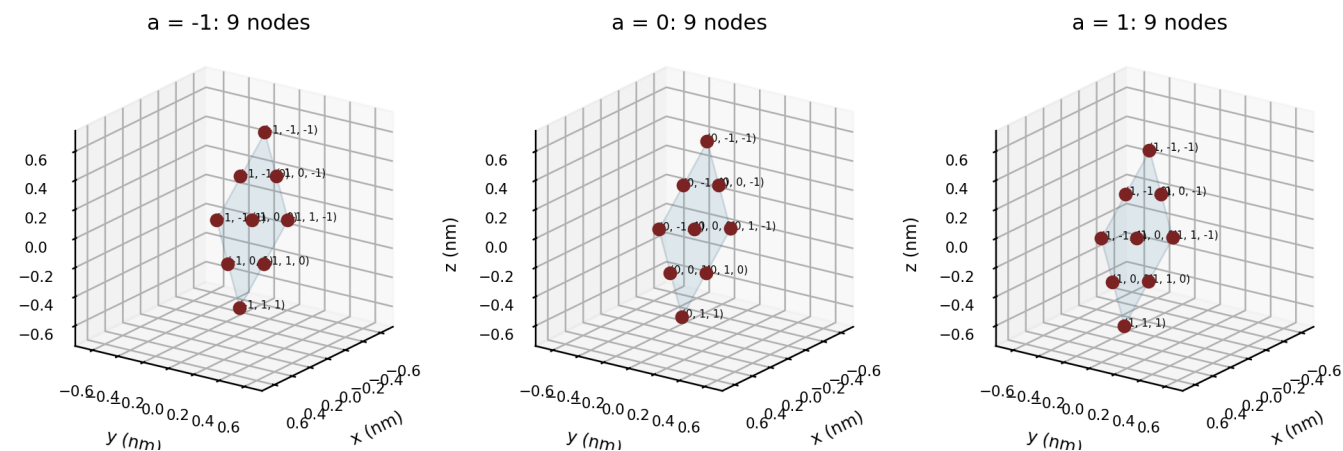


Figure 4. All three centered half-step planes for channel a, parallel to the bc face family. Each panel contains the complete nine-node plane membership at true relative scale.

Cell3 plane family b: fixed-b planes parallel to face ac
Adjacent centered planes = 0.225798689862 nm; primitive spacing = 0.451597379724 nm

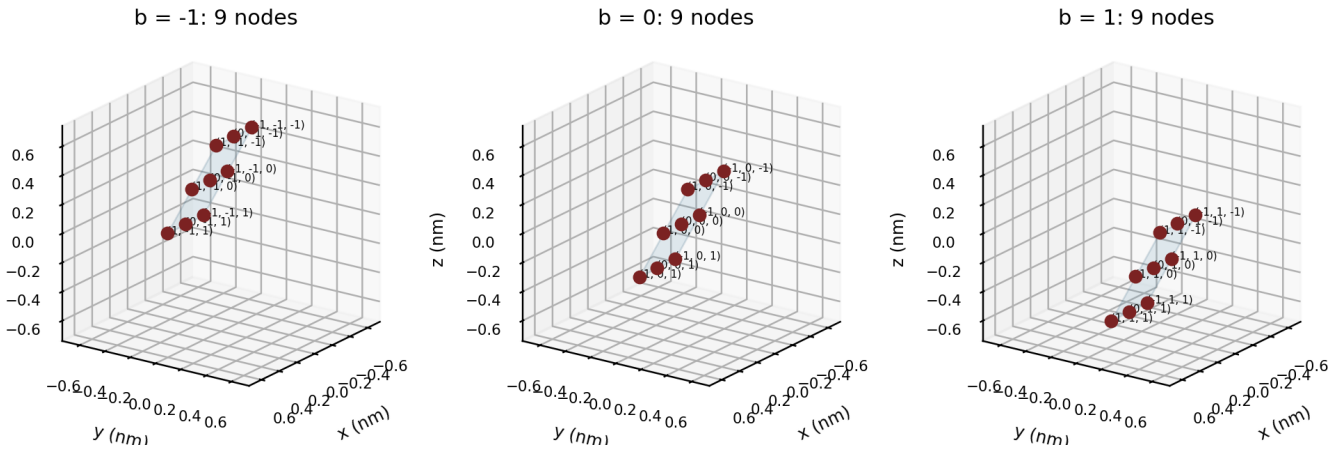


Figure 5. All three centered half-step planes for channel b, parallel to the ac face family. Each panel contains the complete nine-node plane membership at true relative scale.

Cell3 plane family c: fixed-c planes parallel to face ab
Adjacent centered planes = 0.302548870757 nm; primitive spacing = 0.605097741515 nm

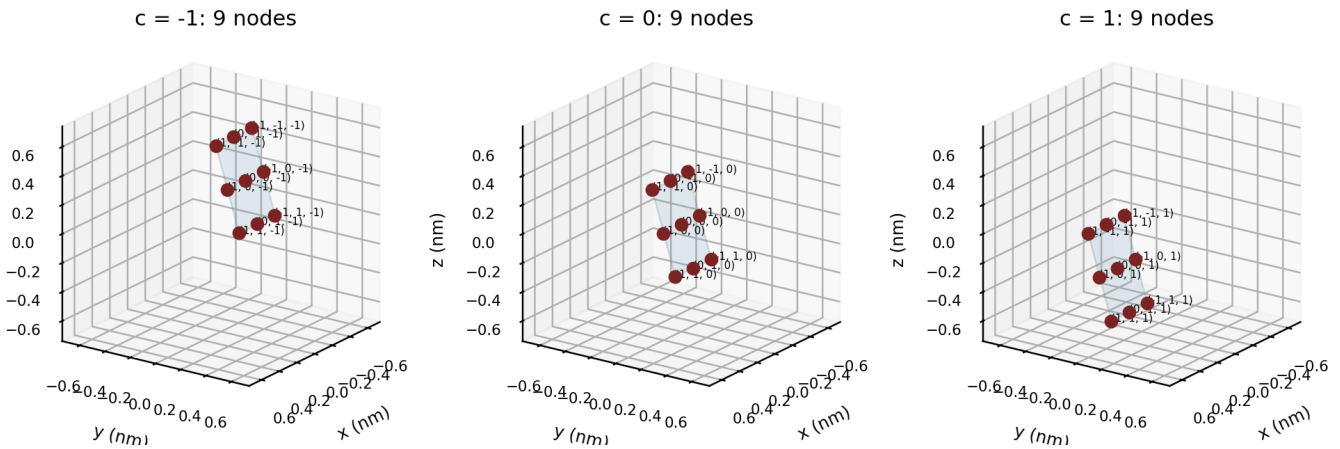


Figure 6. All three centered half-step planes for channel c, parallel to the ab face family. Each panel contains the complete nine-node plane membership at true relative scale.

3. Strict Linear Inertia and the Earth-CMB Mixed State

3.1 Rate-state kinematics

The supplied linear-inertia construction maps every physical velocity into a unique three-channel Cell3 traversal rate. The raw phase scalar is quadratic in that rate, and the direction-dependent normalization recovers ordinary kinetic energy per unit mass. Linear inertia is therefore represented as resistance to changing a coherent traversal-rate state.

$$\begin{aligned} \dot{q} &= 2 \mathbf{B}^T(-1) \mathbf{v} \\ \Phi &= \dot{q}^T \mathbf{W} \dot{q} = \mathbf{v}^T \mathbf{G}_{\text{phase}} \mathbf{v} \\ E/m &= \mathbf{v}^2/2 \end{aligned}$$

For local motion $\mathbf{v}_R$ superposed on the Earth-CMB translation $\mathbf{V}_E$, the raw phase state expands into the Earth baseline, the local self-term, and the mixed interference term.

$$\begin{aligned} \Phi_{\text{total}} &= \Phi_E + \Phi_R + 2 \dot{q}_R^T \mathbf{W} \dot{q}_E \\ \epsilon_{\text{cross}} &= \kappa_F \dot{q}_R^T \mathbf{W} \dot{q}_E = \kappa_F \mathbf{v}_R^T \mathbf{G}_{\text{phase}} \mathbf{V}_E \end{aligned}$$

3.2 Earth-CMB vectors and source coefficient

Quantity	Vector or value	Unit
V_E	(-359003.199065813038, 76679.886034001989, -44701.060100687995)	m/s
$ V_E $	369812.502000406443	m/s
$\dot{q}_{dot_E}$	(-2.217734842862e+15, 2.499065863112e+14, 5.181751775713e+13)	1/s
$g = G_{phase} V_E$	(-880221.981635091361, -143911.436561599461, 266335.026948619285)	m/s
$ g $	930825.217270434834	m/s
κ_F	3.074050899702035e-11	dimensionless source coefficient
$\pi \kappa_F g $	0.000089893660889	m/s ² per Hz

Earth-CMB traversal, phase-gradient direction, optimized plate normal, and lateral output

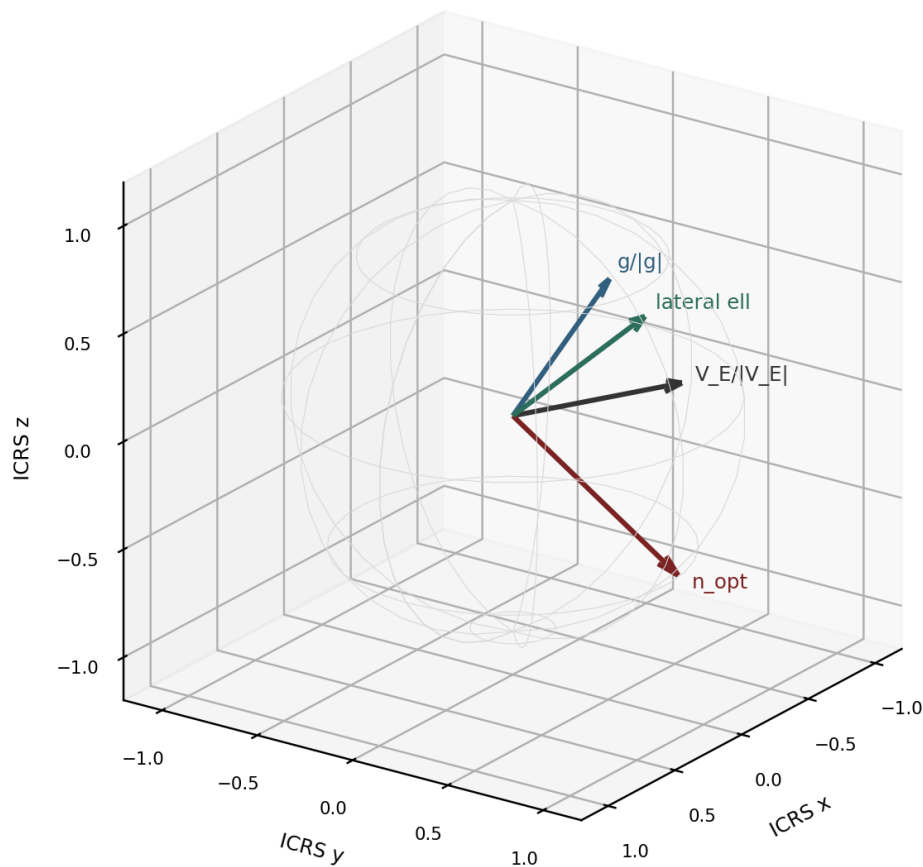


Figure 7. The strict orientation triad derived from the supplied Earth-CMB velocity and G_{phase} . The plate normal n_{opt} is perpendicular to g and lies in the V_E - g plane; the output direction $ell = n_{opt} \times \hat{g}$ is perpendicular to V_E .

4. Antisymmetric Configuration-Space Current

A single reciprocal coordinate closes on one line and has no antisymmetric area. Two nonparallel coordinates with a phase shift create a circulation in their shared configuration space. For plate coordinates x and y , the cycle-averaged current is the signed area rate J_{xy} .

$$J_{xy} = \langle x \dot{y} - y \dot{x} \rangle$$

For $x = A_x \cos(\omega t + \psi)$, $y = A_y \cos(\omega t + \psi - \delta)$:

$$J_{xy} = A_x A_y \omega \sin(\delta)$$

The physical current vector is obtained by attaching the two coordinate axes to the plate plane. With $e_1 \times e_2 = n$, the current becomes $\Omega = J_{xy} n$. Its sign is set by the internal phase order. A common carrier phase ψ does not change J_{xy} .

$$\begin{aligned}\Omega &= \langle \mathbf{r} \times \dot{\mathbf{r}} \rangle = J_{xy} \, n \\ \rho_c^2 &= \langle \mathbf{r}^2 \rangle = (A_x^2 + A_y^2)/2 \\ C_{pol} &= 2 A_x A_y \sin(\delta) / (A_x^2 + A_y^2)\end{aligned}$$

C_{pol} is completely measurable. Equal amplitudes with a 90-degree phase offset give $C_{pol} = +1$ or -1 . Linear motion, zero phase separation, or 180-degree separation gives $C_{pol} = 0$. Amplitude mismatch and quadrature error reduce the magnitude continuously.

$$\begin{aligned}a_K &= [\kappa_F / (2 \rho_c^2)] \, \Omega \times \mathbf{g} \\ &= \pi \kappa_F f C_{pol} (n \times \mathbf{g})\end{aligned}$$

5. Why One Two-Axis Plate Replaces the Thirteen-Axis Synthesis

The thirteen-axis construction distributes one-dimensional coordinates around a plane and phase-locks them so that their coherent first moment is a circle. The direct plate module starts with the two independent coordinates of that circle. Both constructions reduce to the same rank-two physical displacement and the same antisymmetric bivector.

$$\begin{aligned}\text{Thirteen-axis synthesis after normalization:} \\ \mathbf{r}_{13}(t) &= A [\cos(\omega t) \, \mathbf{e}_1 + \sin(\omega t) \, \mathbf{e}_2] \\ \text{Direct plate module:} \\ \mathbf{r}_{plate}(t) &= A [\cos(\omega t) \, \mathbf{e}_1 + \sin(\omega t) \, \mathbf{e}_2]\end{aligned}$$

The strict local composition law depends on $\Omega = \langle \mathbf{r} \times \dot{\mathbf{r}} \rangle$ and ρ_c^2 , so it does not retain the number of scalar actuator channels used to produce the same coherent two-dimensional state. The plate is therefore the minimal direct implementation: two linear drives, one moving mass, and one controlled chirality.

6. One Planar Mass Module

6.1 Geometry and mass

Specified 100 mm x 100 mm x 1 mm plate and its two-axis center-of-mass motion

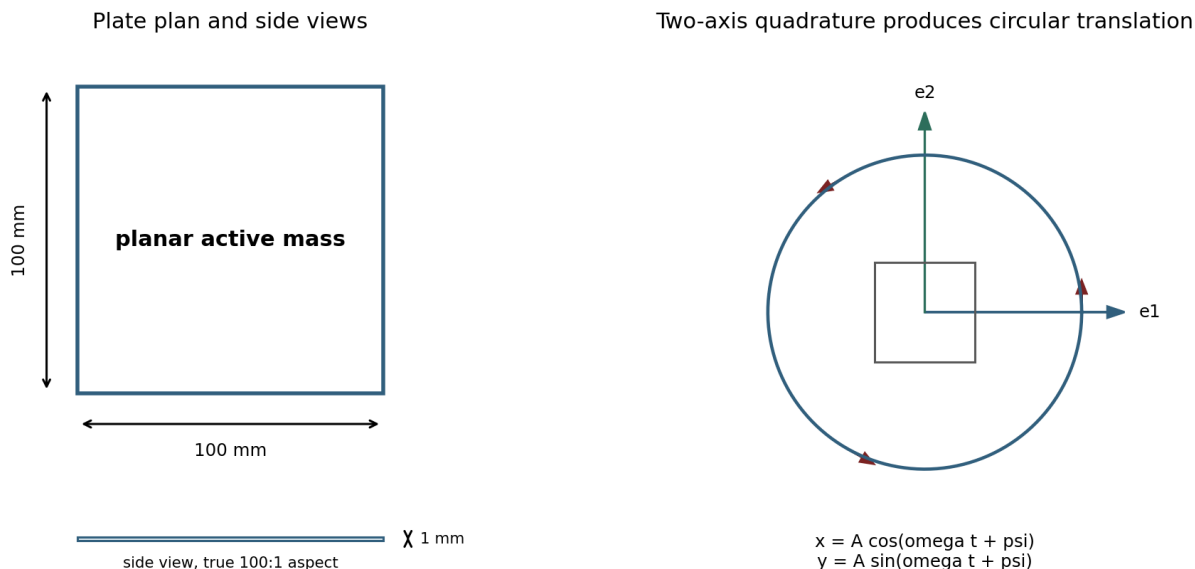


Figure 8. Dimensioned plate geometry and the direct two-coordinate circular translation. The plate orientation remains fixed; only its center of mass translates in the $\mathbf{e}_1$ - $\mathbf{e}_2$ plane.

Parameter	Expression	Design value
Length	L	0.100 m
Width	W	0.100 m
Thickness	t	0.001 m
Volume	$L W t$	0.00001 m ³

Parameter	Expression	Design value
Density input	ρ_m	1600 kg/m ³
Mass per plate	$m_p = \rho_m L W t$	0.016 kg
Three-plate active mass	$3 m_p$	0.048 kg

The value 1600 kg/m³ is the explicit density input used for the numerical carbon-composite design evaluation. The strict acceleration per active kilogram does not depend on material. Material changes total force through moving mass and determines whether the required rigid translation, mode matching, damping, thermal stability, and stress margin can be achieved.

6.2 Two-axis drive law

$$\begin{aligned} r_k(t) = & A_x \cos(\omega t + \psi_k) e_1 \\ & + A_y \cos(\omega t + \psi_k - \delta) e_2 \end{aligned}$$

The plate must be actuated through its center of mass with symmetric flexures or equivalent constraints so that the commanded motion is translation rather than plate rotation or bending. The two actuator channels must share one frequency. Their amplitude ratio and phase difference determine C_{pol} directly.

$$\begin{aligned} \text{Circular case: } & A_x = A_y = A, \delta = +90 \text{ degrees} \\ r_k(t) = & A [\cos(\omega t + \psi_k) e_1 + \sin(\omega t + \psi_k) e_2] \end{aligned}$$

The plate orientation remains stationary. Its center of mass traces a circle of radius A about its equilibrium point. Reversing δ from +90 to -90 degrees reverses the circulation and the calculated lateral force without changing the RMS displacement or carrier frequency.

7. Optimum Plate Orientation

The magnitude of $n \times g$ is maximized when the plate normal is perpendicular to g . To make the force lateral to the Earth-CMB translation, n is selected as the normalized component of V_E perpendicular to g . The in-plane basis is then $e_1 = g/|g|$ and $e_2 = n \times e_1$.

$$\begin{aligned} e_1 &= g/|g| \\ n_{opt} &= \text{normalize}[V_{hat}_E - (V_{hat}_E \cdot e_1) e_1] \\ e_2 &= n_{opt} \times e_1 = \text{normalize}(V_E \times g) \end{aligned}$$

Vector	ICRS Cartesian components	RA (deg)	Dec (deg)	Role
$e_1 = g/ g $	(-0.945636157362, -0.15460629331, 0.28612785946)	189.285391176	16.62627796	first in-plane drive axis
n_{opt}	(-0.315865036407, 0.646162683024, -0.694768354089)	116.050867377	-44.008760899	plate normal
$e_2 = e_1 \times n_{opt}$	(0.077469585422, 0.747375863362, 0.65986951907)	84.082114314	41.289922334	positive lateral output and second drive axis

The angle between V_E and the plate normal is 58.359138729 degrees. The positive lateral direction is perpendicular to V_E to numerical precision. Mounting the plate with the antipodal normal describes the same physical plane; the sign convention is then fixed by the e_1 -to- e_2 drive order.

8. Three Matched Modules and Vibration Cancellation

8.1 Carrier phase assignment

Module	Global carrier phase ψ_k	x-channel phase	y-channel phase for positive chirality
1	0 deg	0 deg	270 deg
2	120 deg	120 deg	30 deg
3	240 deg	240 deg	150 deg

The y channel in this table is 90 degrees behind the corresponding x channel. All three modules therefore have the same chirality. The global 120-degree offsets distribute the modules around the same circular trajectory in time.

$$1 + \exp(i 2 \pi/3) + \exp(i 4 \pi/3) = 0$$

Three-module 120-degree carrier phasing: exact translational cancellation

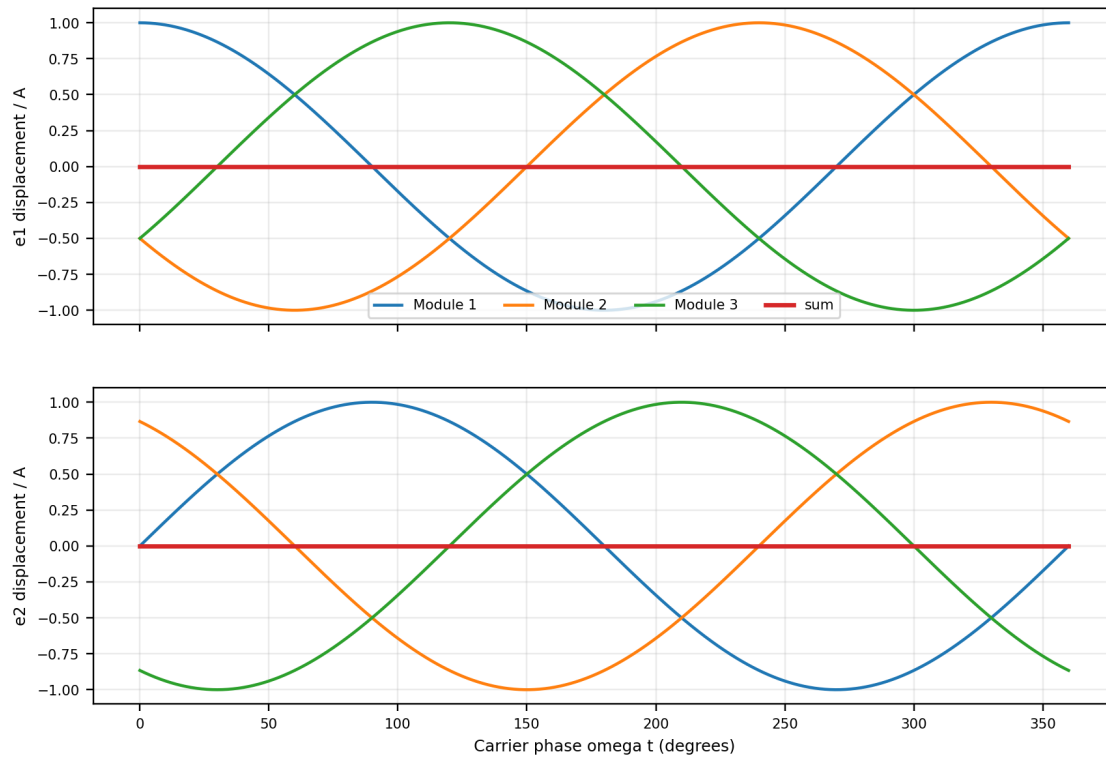


Figure 9. The e1 and e2 displacement components of the three circularly translating modules. At every instant, the three component sums are zero; velocity and acceleration sums therefore vanish as well.

8.2 Ordinary reaction-force cancellation

$$F_{\text{carrier},k}(t) = m_p A \omega^2 [\cos(\omega t + \psi_{i,k}) e_1 + \sin(\omega t + \psi_{i,k}) e_2]$$

$$\sum_k F_{\text{carrier},k}(t) = 0$$

The total moving center of mass is stationary in the ideal matched system. Because the displacement sum is zero at every time, the velocity and acceleration sums are also zero. This removes the first-harmonic rigid-body shaking force from the external support. It does not remove the internal cyclic loads carried separately by each actuator, flexure, plate, and local frame.

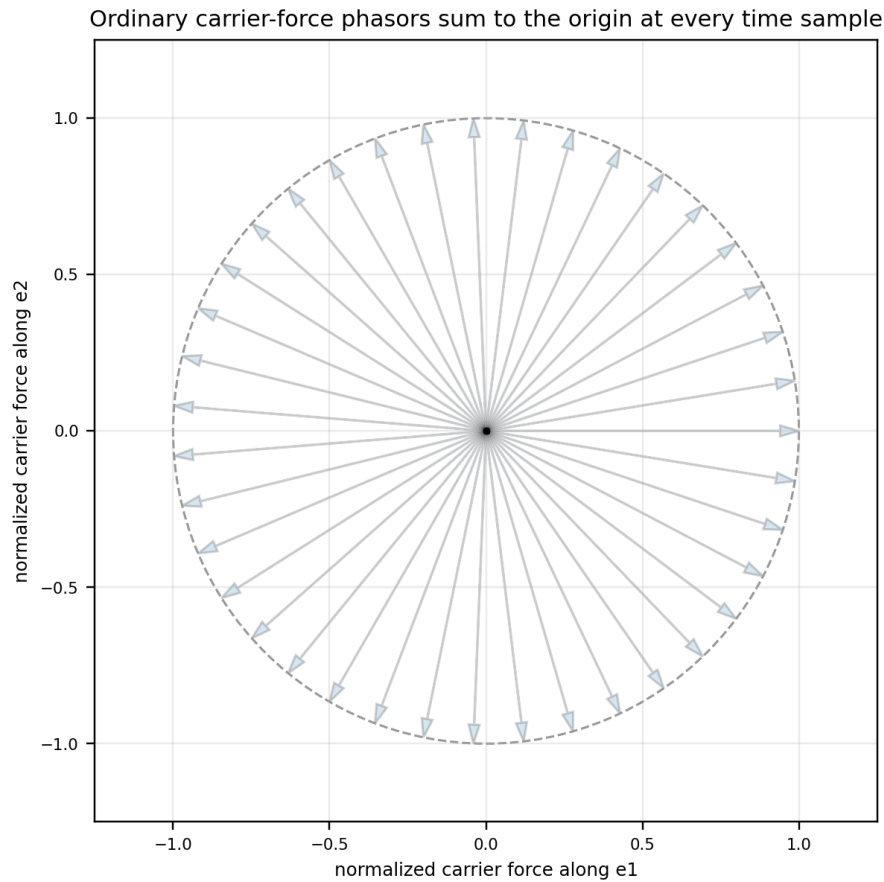


Figure 10. Twelve representative carrier phases are shown. The three individual reaction-force vectors rotate, but their vector sum remains at the origin for ideal matched modules.

8.3 Mean addition and second-harmonic cancellation

The mean lateral term depends on chirality and not on global carrier phase. Each module therefore produces the same mean vector. In contrast, the oscillatory bilinear remainder is at twice the carrier phase. Doubling 0, 120, and 240 degrees produces another complete three-phase set: 0, 240, and 120 degrees.

Mean: $\text{Sum}_k a_{K,\text{mean},k} = 3 a_{K,\text{mean},1}$
 Second harmonic: $\text{Sum}_k \exp(i 2 \psi_k) = 0$

Three-phase separation: 2f ripple cancels while the phase-invariant mean adds

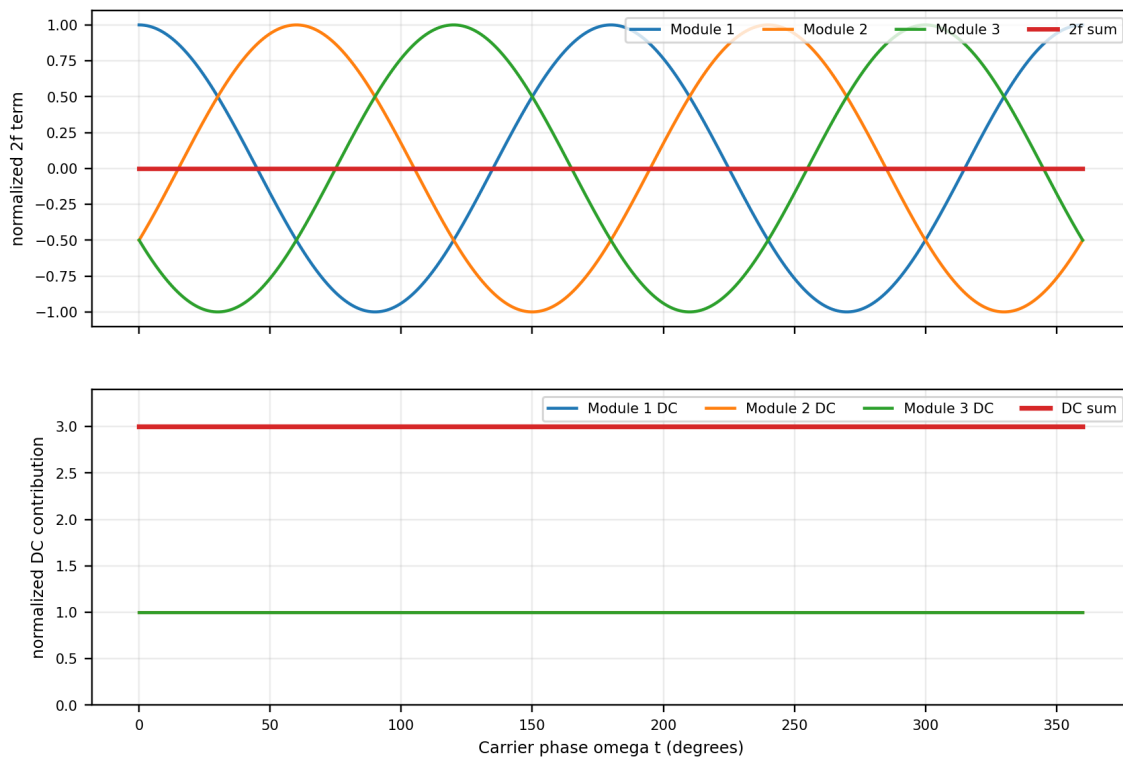


Figure 11. The second-harmonic terms carry phase $2\psi_k$ and sum to zero for $\psi_k = 0, 120, 240$ degrees. The phase-invariant mean contribution is identical in all modules and sums to three.

9. Torque-Balanced Physical Placement

If the three modules are stacked close to a common center, translational cancellation is the dominant requirement. If their centers occupy the vertices of an equilateral triangle, the carrier phase order must be chosen relative to the physical placement azimuth to cancel the net first-harmonic moment.

Physical placement: $\Theta_k = 0, 120, 240$ degrees
Torque-balanced carrier law: $\psi_k = \psi_0 - \Theta_k$

Physical position azimuth Θ_k	Carrier phase ψ_k
0 deg	0 deg
120 deg	240 deg
240 deg	120 deg

The carrier phase progression runs opposite to the spatial placement progression. With this assignment, both the vector force sum and the moment sum about the triangle center vanish in the ideal matched model. The three mean lateral forces are parallel and equal; because the position vectors sum to zero, their mean moment about the geometric center also vanishes.

Sum_k $R_k = 0$
Sum_k $F_{\text{carrier},k} = 0$
Sum_k $R_k \times F_{\text{carrier},k} = 0$
Sum_k $R_k \times F_{K,\text{mean},k} = 0$

Start-up and shut-down ramps must preserve the three-phase relationship. The circulating active masses store orbital angular momentum about their local equilibrium centers during steady operation; symmetric ramps prevent avoidable transient reaction impulses.

10. Instantaneous Response of One Optimally Oriented Circular Module

For the optimized basis, e_1 is parallel to g and e_2 is perpendicular to g . Let $\theta = \omega t + \psi$. The local velocity projection is $\dot{g} = -A \omega |g| \sin(\theta)$. Substitution into the strict circular coordinate-differential kernel separates the response into a constant term and a second-harmonic term.

$$a_k(t) = [\kappa_F \omega |g|/2] \{ \sin(2\theta) e_1 + [1 - \cos(2\theta)] e_2 \}$$

The cycle mean is $\kappa_F \omega |g| e_2 / 2 = \pi \kappa_F f |g| e_2$. The oscillatory components are proportional to $\sin(2\theta)$ and $\cos(2\theta)$. This explicit decomposition explains why the three-module carrier phase set cancels the $2f$ terms while retaining the mean.

11. Force and Acceleration Scaling

$$a_{K,max} = \pi \kappa_F |g| f C_{pol}$$

$$F_{K,three} = 3 m_p a_{K,max}$$

At full circularity and optimum orientation, $\pi \kappa_F |g| = 8.98936608879062e-05 \text{ m/s}^2 \text{ per hertz}$. For three 0.016 kg plates, the total active mass is 0.048 kg and the force coefficient is $4.314895722661952e-06 \text{ N per hertz}$.

Frequency	Acceleration	Three-plate force
1000 Hz	0.089893661 m/s ²	0.004314896 N
10000 Hz	0.898936609 m/s ²	0.043148957 N
100000 Hz	8.989366089 m/s ²	0.431489572 N

Strict linear-inertia scaling for three 16 g plates at full circularity

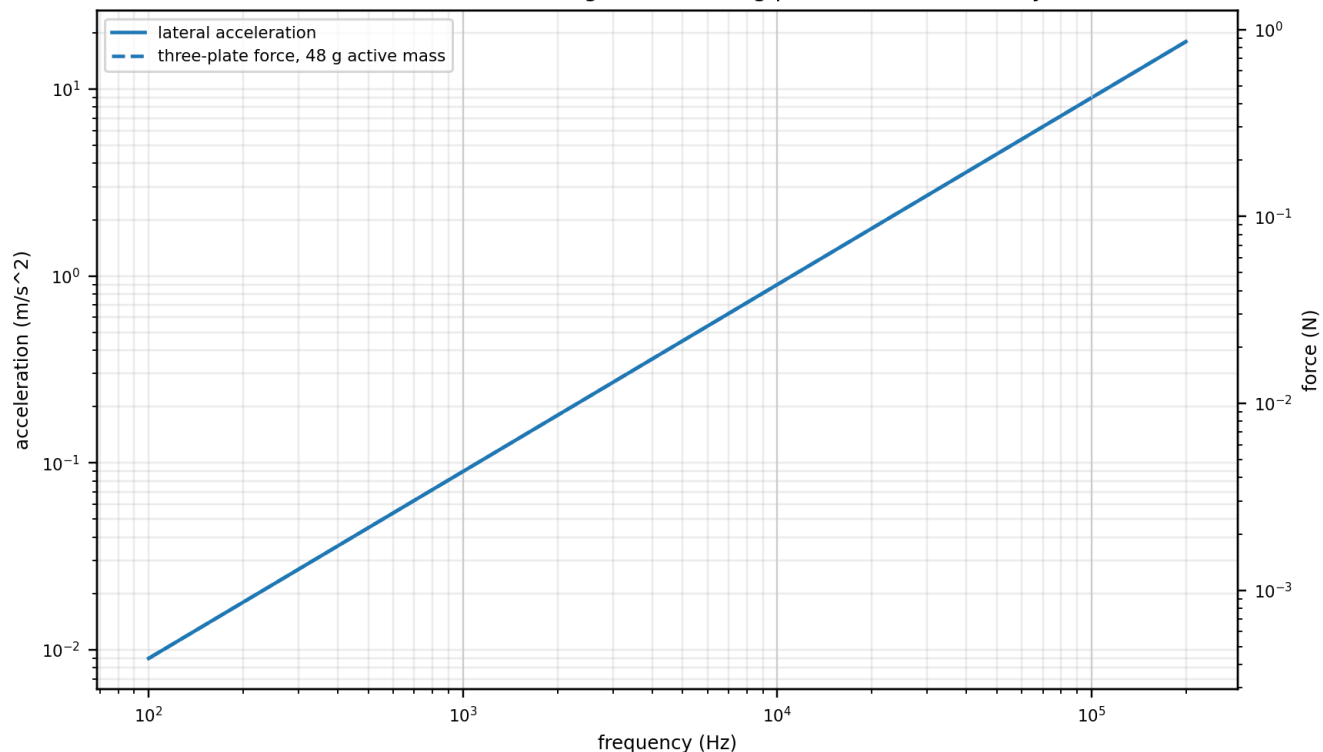


Figure 12. Deterministic evaluation of $|a_K| = \pi \kappa_F |g| f$ and $F_K = M_{active} |a_K|$ for the 48 g three-plate design value. The field equation is linear in frequency and contains no finite internal optimum.

The calculated acceleration equals one standard gravity, 9.80665 m/s^2 , at approximately 109.092 kHz for $C_{pol} = 1$ and optimum orientation. This value is a strict equation scaling point. The plate mechanics determine whether the required coherent rigid-translation mode can be maintained at that frequency.

11.1 Arbitrary density and circularity

$$m_p = 1.0e-5 \rho_m$$

$$F_{K,three} = 2.696809826664e-9 \rho_m f C_{pol} N$$

Density increases total force at fixed geometry because it increases the active mass. The acceleration remains the same for every coherently participating kilogram. A lower-density high-specific-stiffness plate may permit a cleaner or higher-frequency rigid mode, while a denser plate provides more force but demands proportionally larger carrier forces from the actuators.

12. Structural Loads, Mismatch, and Balance Tolerance

$$F_{carrier,peak} = m_p A (2 \pi f)^2$$

For one 0.016 kg plate at $A = 1$ micrometre and $f = 10$ kHz, the peak ordinary inertial force is 63.165468 N. The ideal vector sum of the three carrier forces is zero, but each local actuator and flexure still carries this cyclic load.

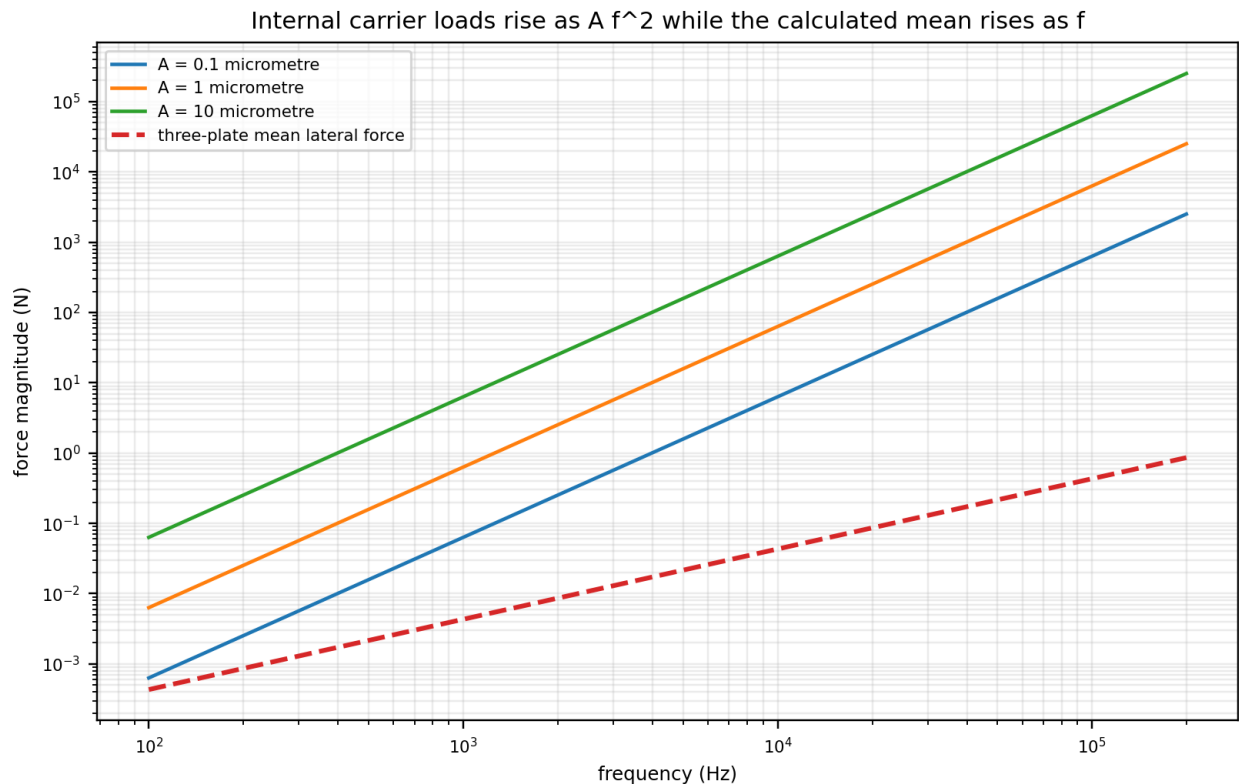


Figure 13. Peak ordinary inertial force for one 16 g plate at three commanded translation amplitudes, compared with the calculated total mean force of all three plates. Ideal vector cancellation does not remove these internal cyclic loads from actuators and flexures.

12.1 Approximate phasor mismatch criterion

Represent the residual first-harmonic carrier imbalance by a fractional phasor error epsilon relative to one plate carrier force. Requiring the raw carrier mismatch to remain below the calculated total mean force gives the following approximate design criterion.

$$\epsilon < 6.8310991e-6 / (A f)$$

A in metres; f in hertz

At $A = 1$ micrometre and $f = 10$ kHz, epsilon must be below 0.068311 percent. Interpreted as a single small phase error, this corresponds to approximately 0.039139 degree. Frequency-domain separation helps distinguish a DC channel from a 1f mismatch, but structural nonlinearities can mix carrier vibration into low-frequency signals; active amplitude and phase trimming remain necessary.

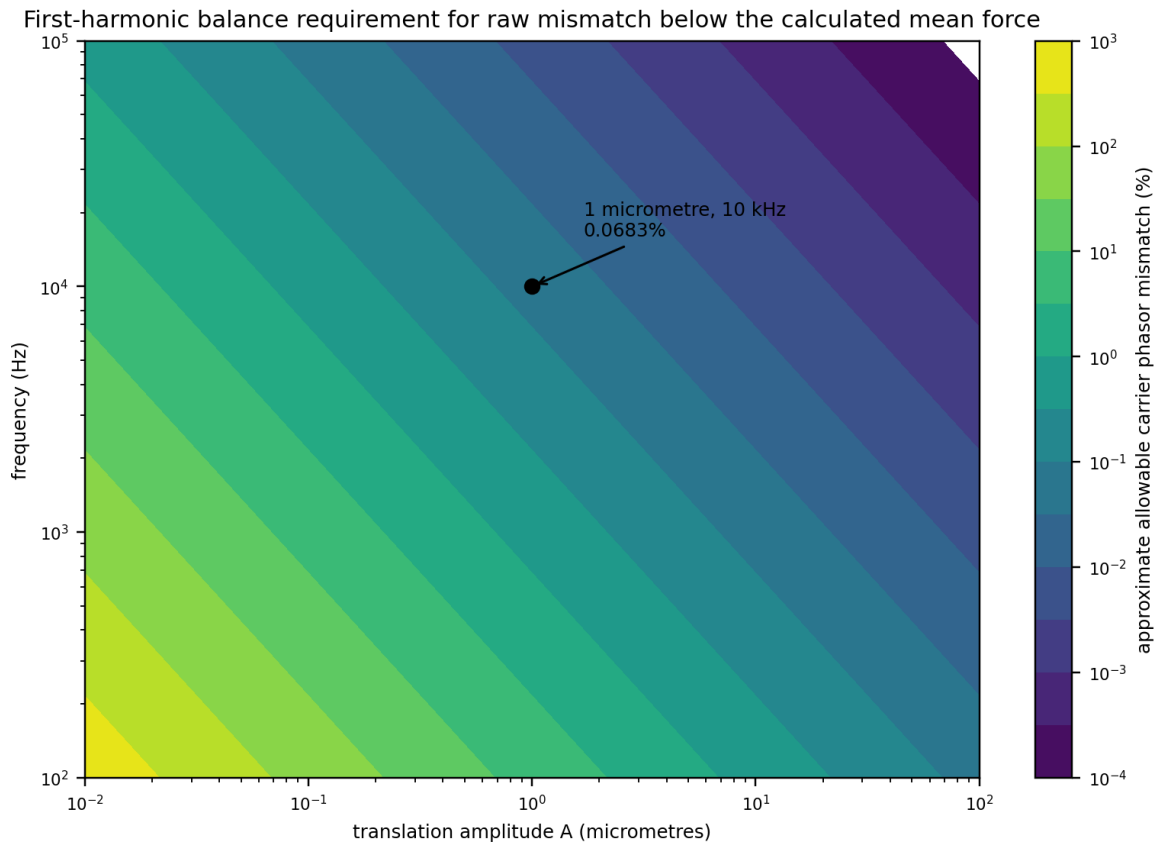


Figure 14. Approximate allowable residual carrier-force phasor mismatch $\epsilon < 6.8310991e-6/(A f)$, with A in metres. This is a structural balancing criterion, not an additional K-field coefficient.

13. Frequency and Material Selection

The strict field equation is linear in frequency and contains no finite interior optimum. Frequency is therefore selected by the mechanical implementation. The correct operating point is the highest frequency for which the intended effective plate mass executes matched two-axis translation with stable circularity and acceptable stress, heating, and control error.

```
f_operating = highest f satisfying:
matched orthogonal translation modes
stable A_x/A_y and delta
C_pol near the commanded value
effective moving mass known
structural stress and actuator force limits satisfied
phase control and thermal stability satisfied
```

An ultrastiff quasi-isotropic carbon-fiber laminate is attractive when high specific stiffness and low mass improve modal separation. A strongly directional laminate can split the two in-plane responses and complicate quadrature matching. An isotropic ceramic or metal plate can simplify modal matching but increases moving mass and carrier load. The report therefore treats material through measured density, stiffness matrix, damping, mode shapes, thermal coefficients, and actuator coupling rather than through an unknown K-field transfer factor.

The two drive axes should be calibrated as a complex 2x2 transfer matrix. Command voltages are then adjusted so the measured center-of-mass motion has equal orthogonal amplitudes and the selected quadrature phase. This direct measurement determines C_pol and the participating mass at the chosen frequency.

14. Control Architecture and Experimental Observables

14.1 Six synchronized actuator channels

Three plates require six synchronized motion channels: x and y for each module. A common clock should generate all phases. Each plate receives a closed-loop two-axis correction based on displacement sensing, while a slower supervisory loop trims the three global carrier phases and amplitudes to minimize frame acceleration at 1f and 2f.

Control quantity	Target	Diagnostic
Internal amplitude ratio A_y/A_x	1.000	measured plate-center orbit ellipse
Internal phase delta	+90 deg or -90 deg	chirality and C_{pol}
Global carrier separation	120 deg	frame 1f acceleration null
Doubled phase separation	120 deg after modulo 360	frame 2f response null
Physical placement phase law	$\psi_k = \psi_0 - \Theta_k$	frame moment null
Common chirality	same sign for all modules	mean lateral addition

14.2 Required reversals and controls

Chirality reversal. Change the internal phase from +90 to -90 degrees in all modules. The mean lateral vector reverses while RMS motion and carrier frequency remain unchanged.

Carrier phase rotation. Add the same phase to both axes of one module. This changes its instantaneous point on the orbit but not its mean contribution.

Linear-polarization control. Set delta to 0 or 180 degrees while holding comparable RMS displacement and electrical loading. C_{pol} becomes zero.

Orientation sweep. Rotate the common plate normal relative to g. The magnitude follows $|n \times g|$ and vanishes when n is parallel to g.

Module omission. Operate one, two, and three plates to separate single-module response from balancing performance.

15. Consolidated Design Specification

Item	Specified result
Active module	Rigid 100 mm x 100 mm x 1 mm planar mass with two orthogonal in-plane translation drives
Module count	Three matched modules
Internal drive	Equal-frequency x and y motion; equal amplitudes for circular operation; internal phase +/-90 degrees
Common chirality	All modules use the same internal phase order
Global phase set	0, 120, 240 degrees
Torque-balanced placement law	At physical azimuth Θ_k , use $\psi_k = \psi_0 - \Theta_k$
Plate normal n_{opt}	(-0.315865036407, 0.646162683024, -0.694768354089)
Drive axis e1	$g/ g = (-0.945636157362, -0.154606293310, 0.286127859460)$
Drive axis e2 / positive output	(0.077469585422, 0.747375863362, 0.659869519070)
Strict acceleration law	$a_K = \pi \kappa_F f C_{pol} (n \times g)$
Maximum coefficient	8.989366088879e-5 m/s ² per Hz per active kilogram
Design density evaluation	1600 kg/m ³ ; 0.016 kg per plate; 0.048 kg total active mass
Three-plate force coefficient	4.314895722662e-6 N/Hz at $C_{pol} = 1$ and optimum orientation
Unknown field factor	None
Frequency selection	Highest mechanically valid matched circular-translation mode: no finite intrinsic maximum in the strict law

Final result. One two-axis plate supplies the strict antisymmetric current. Three identical plates, separated by 120 degrees of global carrier phase and operated with the same internal chirality, cancel the ideal structural 1f force, cancel the ideal 2f ripple, and add the calculated mean lateral force. The thirteen-axis array is not required for the leading strict-linear effect.

Appendix A. Complete 27-Node Cell3 Ledger

Address	Kind	lambda	Polarity	x (nm)	y (nm)	z (nm)
(-1, -1, -1)	corner	-3	-1	-0.176351350143518	0.118682429904309	0.637311498325592
(-1, -1, 0)	edge center	-2	1	-0.181416008956851	-0.076211421277593	0.292444294144131
(-1, 0, -1)	edge center	-2	1	-0.158685962221493	0.222342355125059	0.363610955629613
(0, -1, -1)	edge center	-2	1	-0.012600729108693	0.091233925961153	0.61856774687744
(-1, -1, 1)	corner	-1	-1	-0.186480667770183	-0.271105272459495	-0.05242291003733
(-1, 0, 0)	face center	-1	-1	-0.163750621034825	0.027448503943156	0.018743751448153
(-1, 1, -1)	corner	-1	-1	-0.141020574299467	0.326002280345807	0.089910412933635
(0, -1, 0)	face center	-1	-1	-0.017665387922025	-0.103659925220749	0.273700542695978
(0, 0, -1)	face center	-1	-1	0.005064658813332	0.194893851181902	0.344867204181461
(1, -1, -1)	corner	-1	-1	0.151149891926133	0.063785422017997	0.599823995429287
(-1, 0, 1)	edge center	0	1	-0.168815279848158	-0.167445347238745	-0.326123452733309
(-1, 1, 0)	edge center	0	1	-0.1460852331128	0.131108429163905	-0.254956791247826
(0, -1, 1)	edge center	0	1	-0.022730046735358	-0.298553776402651	-0.071166661485483
(0, 0, 0)	body center	0	1	0	0	0
(0, 1, -1)	edge center	0	1	0.022730046735358	0.298553776402651	0.071166661485483
(1, -1, 0)	edge center	0	1	0.1460852331128	-0.131108429163905	0.254956791247826
(1, 0, -1)	edge center	0	1	0.168815279848158	0.167445347238745	0.326123452733309
(-1, 1, 1)	corner	1	-1	-0.151149891926133	-0.063785422017997	-0.599823995429287
(0, 0, 1)	face center	1	-1	-0.005064658813332	-0.194893851181902	-0.344867204181461
(0, 1, 0)	face center	1	-1	0.017665387922025	0.103659925220749	-0.273700542695978
(1, -1, 1)	corner	1	-1	0.141020574299467	-0.326002280345807	-0.089910412933635
(1, 0, 0)	face center	1	-1	0.163750621034825	-0.027448503943156	-0.018743751448153
(1, 1, -1)	corner	1	-1	0.186480667770183	0.271105272459495	0.05242291003733
(0, 1, 1)	edge center	2	1	0.012600729108693	-0.091233925961153	-0.61856774687744
(1, 0, 1)	edge center	2	1	0.158685962221493	-0.222342355125059	-0.363610955629613
(1, 1, 0)	edge center	2	1	0.181416008956851	0.076211421277593	-0.292444294144131
(1, 1, 1)	corner	3	-1	0.176351350143518	-0.118682429904309	-0.637311498325592

Appendix B. Complete Plane Membership Ledger

Channel a: planes parallel to face bc

Plane address	Node 1	Node 2	Node 3	Node 4	Node 5	Node 6	Node 7	Node 8	Node 9
-1	(-1, -1, -1)	(-1, -1, 0)	(-1, -1, 1)	(-1, 0, -1)	(-1, 0, 0)	(-1, 0, 1)	(-1, 1, -1)	(-1, 1, 0)	(-1, 1, 1)
0	(0, -1, -1)	(0, -1, 0)	(0, -1, 1)	(0, 0, -1)	(0, 0, 0)	(0, 0, 1)	(0, 1, -1)	(0, 1, 0)	(0, 1, 1)
1	(1, -1, -1)	(1, -1, 0)	(1, -1, 1)	(1, 0, -1)	(1, 0, 0)	(1, 0, 1)	(1, 1, -1)	(1, 1, 0)	(1, 1, 1)
Quantity				Value					
Unit normal				(-0.995965190308961, 0.083602162507377, -0.0326192905644)					
Adjacent centered half-step spacing				0.164773264854647 nm					
Primitive boundary spacing				0.329546529709293 nm					
Face area				0.357809539128562 nm ²					

Channel b: planes parallel to face ac

Plane address	Node 1	Node 2	Node 3	Node 4	Node 5	Node 6	Node 7	Node 8	Node 9
-1	(-1, -1, -1)	(-1, -1, 0)	(-1, -1, 1)	(0, -1, -1)	(0, -1, 0)	(0, -1, 1)	(1, -1, -1)	(1, -1, 0)	(1, -1, 1)
0	(-1, 0, -1)	(-1, 0, 0)	(-1, 0, 1)	(0, 0, -1)	(0, 0, 0)	(0, 0, 1)	(1, 0, -1)	(1, 0, 0)	(1, 0, 1)
1	(-1, 1, -1)	(-1, 1, 0)	(-1, 1, 1)	(0, 1, -1)	(0, 1, 0)	(0, 1, 1)	(1, 1, -1)	(1, 1, 0)	(1, 1, 1)
Quantity				Value					
Unit normal				(0.089052594101605, 0.866576769291032, -0.491033948326276)					
Adjacent centered half-step spacing				0.225798689862065 nm					
Primitive boundary spacing				0.45159737972413 nm					
Face area				0.261106235799531 nm ²					

Channel c: planes parallel to face ab

Plane address	Node 1	Node 2	Node 3	Node 4	Node 5	Node 6	Node 7	Node 8	Node 9
-1	(-1, -1, -1)	(-1, 0, -1)	(-1, 1, -1)	(0, -1, -1)	(0, 0, -1)	(0, 1, -1)	(1, -1, -1)	(1, 0, -1)	(1, 1, -1)
0	(-1, -1, 0)	(-1, 0, 0)	(-1, 1, 0)	(0, -1, 0)	(0, 0, 0)	(0, 1, 0)	(1, -1, 0)	(1, 0, 0)	(1, 1, 0)
1	(-1, -1, 1)	(-1, 0, 1)	(-1, 1, 1)	(0, -1, 1)	(0, 0, 1)	(0, 1, 1)	(1, -1, 1)	(1, 0, 1)	(1, 1, 1)
Quantity				Value					
Unit normal				(0.194092200807767, 0.913177168821229, 0.358379234790805)					
Adjacent centered half-step spacing				0.302548870757372 nm					
Primitive boundary spacing				0.605097741514743 nm					
Face area				0.1948691654699 nm^2					

Appendix C. Matrix and Vector Ledger

B basis

Row	Column 1	Column 2	Column 3
1	0.327501242069651	0.035330775844051	-0.010129317626665
2	-0.054897007886313	0.207319850441498	-0.389787702363804
3	-0.037487502896305	-0.547401085391957	-0.689734408362922

Unit: nm.

M = B^T B

Row	Column 1	Column 2	Column 3
1	0.1116760579054345	0.0207103332816065	0.043937235097789
2	0.0207103332816065	0.3438777323971203	0.2963927589615229
3	0.043937235097789	0.2963927589615229	0.6277706100693854

Unit: nm^2.

R = M^(-1)

Row	Column 1	Column 2	Column 3
1	9.2080255286837733	0.0015348024281183	-0.6451880356768953
2	0.0015348024281183	4.903398312306197	-2.3151755844163877
3	-0.6451880356768953	-2.3151755844163877	2.7311712747461088

Unit: nm^-2.

W address coupling

Row	Column 1	Column 2	Column 3
1	0.0833533026334571	0.1375005766035881	0.1025592112219582
2	0.1375005766035881	0.6924000428485476	0.5508009044534695
3	0.1025592112219582	0.5508009044534695	0.5877476523533913

Unit: nm^2.

G_phase

Row	Column 1	Column 2	Column 3
1	2.9764106377999999	1.2431507795000001	-2.0803600354
2	1.2431507795000001	2.5922286634999998	-2.3178839937000002
3	-2.0803600354	-2.3178839937000002	6.7735709133000004

Unit: dimensionless.

Appendix D. Detailed Three-Phase Identities

Let $z_k = \exp(i \psi_k)$ with $\psi_k = 0, 2\pi/3, 4\pi/3$. The roots-of-unity identity gives $z_1 + z_2 + z_3 = 0$. Differentiation multiplies every phasor by the same $i\omega$ factor, so displacement, velocity, acceleration, and ordinary inertial-force sums all vanish.

$$\begin{aligned}\text{Sum } z_k &= 0 \\ \text{Sum } i\omega z_k &= 0 \\ \text{Sum } -\omega^2 z_k &= 0\end{aligned}$$

The second harmonic carries z_k^2 . Squaring permutes the two nontrivial cube roots, so their sum also vanishes. The mean term has no z_k factor and therefore sums to three.

$$\begin{aligned}\text{Sum } z_k^2 &= 0 \\ \text{Sum } 1 &= 3\end{aligned}$$

For an equilateral physical placement $R_k = R \exp(i \Theta_k)$, use $\psi_k = \psi_0 - \Theta_k$. The carrier moment is proportional to the imaginary part of $\text{Sum} \exp[i(\psi_k - \Theta_k)]$, which becomes a three-root sum at twice the spatial azimuth and vanishes.

Appendix E. Dimensional Closure

Object	Dimensions
κ_F	dimensionless in the supplied normalized law
$g = G_{\text{phase}} V_E$	m/s
f	1/s
$\pi \kappa_F f (n \times g)$	m/s ²
$m_p a_K$	N
$J_{xy} = A_x A_y \omega \sin(\delta)$	m ² /s
$\Omega/(2 \rho_c^2)$	1/s

The plate amplitude cancels from the ideal normalized circular mean because both the oriented current and the coherent radius squared scale as A^2 . Amplitude remains mechanically essential: it determines ordinary carrier load, signal-to-noise conditions, actuator stress, displacement-sensor resolution, and mismatch sensitivity.